

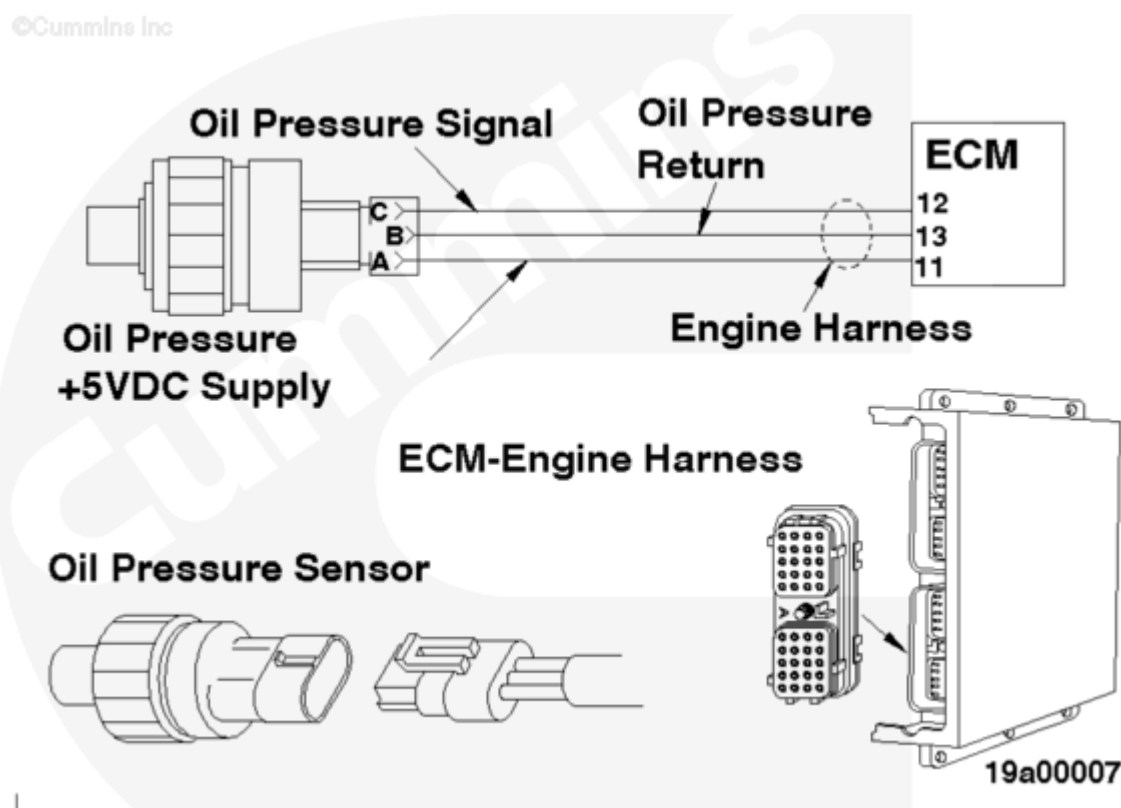
Fault Code: 143

Oil Pressure - Pre-Low Oil Pressure (LOP) Warning

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 143 PID(P): SPN: FMI: Lamp: SRT: 00-354	Engine oil pressure has dropped below the warning threshold for low oil pressure.	No effect on performance. Common Warning output is energized. Pre-Low Oil Pressure (LOP) relay driver is energized.



Circuit Description

The oil pressure sensor is used by the Electronic Control Module (ECM) to monitor the lubricating oil pressure. The ECM monitors the voltage on the signal pin and converts this to a pressure value. The oil pressure value is used by the ECM for the engine protection system.

Component Location

The oil pressure sensor (OPS) is located on the left bank of the engine block above the fuel pump.

Shoptalk

- Confirm that the oil pressure sensor supply voltage is between 4.75 and 5.25 VDC at the sensor. See Fault Code 141.
- Oil pressure is a function of engine speed, oil level, and regulator function. Operating the engine at a low speed under load will **not** cause the oil pressure to be low unless oil is hot, oil at a low level, regulator has malfunctioned, or a loss is occurring somewhere in the system.
- The threshold for the pre-low oil pressure warning is adjustable with INSITE™, Part No. 3825145.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the sensor accuracy.	
	STEP 1A. Verify sensor accuracy with a mechanical gauge.	Sensor reading is correct
STEP 2.	Clear the fault code.	
	STEP 2A. Disable the fault code.	Fault Code 143 inactive
	STEP 2B. Clear any inactive fault codes.	All faults cleared

STEP 1. Check the sensor accuracy.

STEP 1A. Verify sensor accuracy with a mechanical gauge.

Conditions :

- Stop/Run switch in the "STOP" position.
- Controller in the diagnostic mode.

Action	Specification/Repair	Next Step
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• Connect a mechanical oil pressure gauge of known quality and calibration to the engine at one of the plugs on top of the oil filter head. • Connect INSITE™, Part No. 3825145, to the data link. • Start engine and compare the oil pressure reading on the INSITE™, Part No. 3825145, monitor screen to the reading on mechanical oil pressure gauge. NOTE: The engine rpm will need to be increased to make it easier to see differences in the readings.	Sensor reading is correct Refer to the Base Engine Troubleshooting and Repair Manual for correct specifications.	2A
		Go to Fault Code 141

STEP 2. Clear the fault code.

STEP 2A. Disable the fault code.

Conditions : • Connect all components.		
Action	Specification/Repair	Next Step
• Connect all of the components. • Start engine and let it idle for one minute. NOTE: If the fault was at a particular rpm, run engine at that rpm to verify the problem is corrected.	Fault Code 143 inactive Return to the troubleshooting steps or contact your local Cummins Authorized Repair Location if all steps have been completed and rechecked.	2B 1A

STEP 2B. Clear any inactive fault codes.

Conditions :

- Connect all of the components.

Action	Specification/Repair	Next Step
• Erase any inactive fault codes using INSITE™, Part No. 3825145. NOTE: The datalink connector is located on the right bank of the flywheel housing.	All faults cleared	Repair complete
	Troubleshoot any remaining active fault codes.	Appropriate troubleshooting chart

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